Scaling analysis and self-similarity of a nonlinear diffusion equation with $n = 3$

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1 Introduction

This work studies how a fracture aperture (or fluid pressure) propagates along a deformable fracture during fluid injection. The main goal is to determine how the aperture front position depends on time, and how this relationship varies with injection conditions.

Three injection cases are considered: constant pressure, constant flow rate, and a flow rate that increases linearly with time. The fracture is assumed to be semi-infinite. The role of initial conditions is also examined by considering both zero and non-zero initial aperture.

The aperture is governed by the nonlinear diffusion equation (Murphy et al. 2004):

$$M \frac{\partial}{\partial x}\left(w^3 \frac{\partial w}{\partial x}\right) = \frac{\partial w}{\partial t}, \tag{1}$$

where w is the aperture, x is distance, and t is time. The parameter $M = k_n/(12\mu)$, where k_n is stiffness and μ is fluid viscosity.

The aperture front position is expected to follow a power-law in time:

$$x_f \propto t^{\alpha} \tag{2}$$

where x_f is the front position and α is the exponent of time.

Self-similar solutions provide a direct link between space and time in nonlinear diffusion problems and arise under certain simplifying conditions (Barenblatt 1996). Here, self-similarity is used to determine how the front position scales with time, i.e., to find the exponent α, and how it depends on the injection condition.

2 Self-similarity

A time-dependent phenomenon (e.g., aperture, pressure, temperature) is **self-similar** if its spatial distribution changes with time while retaining the same shape (Barenblatt 1996). In this case, there exist time-dependent scales such that, when expressed in these scales, the solution becomes time-independent. As a result, the spatial distribution at different times collapses onto a single curve when written in self-similar variables.

Self-similar solutions typically arise in idealized problems with simple geometry (e.g., infinite or semi-infinite domains) and simple initial and boundary conditions (e.g., an initially unperturbed state). For such problems, finding a self-similar solution provides a complete description of the system.

To construct a self-similar solution, one must first identify appropriate similarity variables. These are obtained from transformations that leave the governing partial differential equation invariant. For the nonlinear diffusion equation (Eq. 1), scaling (or dilation) transformations are particularly relevant (Pakdemirli and Yurusoy 1998; Bluman and Anco 2002).

2.1 Scaling transformation

To find similarity variables, we seek a transformation that leaves the governing PDE (Eq. 1) unchanged. We consider the following **scaling (or dilation) transformation**:

$$\tilde{x} = \varepsilon^a x, \quad \tilde{t} = \varepsilon^b t, \quad \tilde{w} = \varepsilon^c w,$$

where $\varepsilon > 0$ is an arbitrary parameter. Substituting these scaled variables into the PDE allows us to determine the exponents a, b, and c by requiring that the transformed equation matches the original.

For convenience, we first rewrite the PDE as:

$$M\Big(3w^2(w_x)^2 + w^3 w_{xx}\Big) = w_t,$$

where $w_x = \partial w/\partial x$ and $w_{xx} = \partial^2 w/\partial x^2$.

Substituting the scaled variables gives:

$$\varepsilon^{2a-4c} M\left[3\tilde{w}^2(\tilde{w}_{\tilde{x}})^2 + \tilde{w}^3 \tilde{w}_{\tilde{x}\tilde{x}}\right] = \varepsilon^{b-c}\tilde{w}_{\tilde{t}}.$$

For the PDE to be **invariant under scaling**, the powers of ε must match, which leads to:

$$2a - 4c = b - c. \tag{3}$$

This condition ensures that the scaled PDE has the same form as the original.

2.2 Similarity solution

The similarity solution of the nonlinear diffusion equation can be written as

$$w(x,t) = t^{c/b}\,\theta\big(x/t^{a/b}\big),$$

where $\zeta = x/t^{a/b}$ and $\theta(\zeta) = w(x,t)/t^{c/b}$ are the self-similar variables. Substituting this form into the PDE and using the exponent relation from Eq. 3, the PDE successfully reduces to an ODE in $\theta(\zeta)$. The exponents a, b, and c are determined by boundary and initial conditions.

3 Initially closed fracture

For a fracture with zero initial aperture,

$$w(x,0) = 0, \quad x > 0, \qquad w(x,t) \to 0, \quad x \to \infty,$$

in similarity form, this becomes

$$t^{c/b}\theta(\zeta) \to 0, \quad \zeta \to \infty \quad (\text{as } t \to 0),$$

which implies $\theta(\zeta) \to 0$ as $\zeta \to \infty$ and $c/b \geq 0$. The values of a, b, and c depend on the injection boundary condition at $x = 0$. I consider the following **injection conditions**:

- **Constant aperture (pressure):** $w(0,t) = w_0$

  $$t^{c/b}\theta(0) = w_0 \implies c/b = 0$$

  Using $b = 2a - 3c$ (Eq. 3) gives $b = 2a$, leading to

  $$w(x,t) = \theta(\zeta), \quad \zeta = \frac{x}{t^{1/2}}$$

- **Constant injection rate:** $q(0,t) = q_0$

$$t^{(4c-a)/b}\theta(0)^3\theta'(0) = q_0 \implies 4c - a = 0$$

Combined with $b = 2a - 3c$, this gives

$$c/b = 1/5, \quad a/b = 4/5, \quad w(x,t) = t^{1/5}\theta(\zeta), \quad \zeta = x/t^{4/5}$$

- **Linearly increasing injection rate:** $q(0,t) = q_0 t$

$$t^{(4c-a)/b}\theta(0)^3\theta'(0) = q_0 t \implies 4c - a = b$$

Combined with $b = 2a - 3c$, this gives

$$c/b = 3/5, \quad a/b = 7/5, \quad w(x,t) = t^{3/5}\theta(\zeta), \quad \zeta = x/t^{7/5}$$

4 Initially Open Fracture

For a semi-infinite fracture with non-zero initial aperture w_i, the initial condition is

$$w(x,0) = w_i, \quad x > 0, \qquad w(x,t) \to 0, \quad x \to \infty.$$

In similarity variables, this becomes

$$t^{c/b}\theta(\zeta) \to w_i, \quad \zeta \to \infty \quad (\text{as } t \to 0).$$

If $c/b > 0$, the right-hand side depends on time:

$$\theta(\zeta \to \infty) \to \frac{w_i}{t^{c/b}},$$

which violates the self-similarity assumption.

For constant pressure, $c/b = 0$, so the initial aperture does not affect the similarity solution. For constant or linearly increasing injection rates, $c/b > 0$, so the initial aperture prevents a strictly self-similar solution.

However, at **late times** $t \gg t_c$ (where t_c is a critical time), $w_i/t^{c/b} \to 0$, and the similarity solutions derived for zero initial aperture apply. At **early times** $t \ll t_c$, the aperture increase (w) is small compared to w_i, so the fracture behaves almost as rigid, and a linear diffusion solution is appropriate.

The critical time t_c depends on the physical parameters of the problem. The dimensional analysis can be used to find its dependence on the parameters of the problem. For constant injection rate condition the dimensional analysis yields

$$t_c = \frac{k_n w_i^5}{12\mu q_0^2}$$

which shows that, the higher is the applied injection rate, the smaller is the characteristic time.

5 Conclusions

The time exponents α governing aperture propagation derived under all conditions considered are summarized in Table 1.

Table 1: Predicted time exponents α for fracture aperture propagation.

Initial condition	**Injection condition**	**Time regime**	**Exponent α**
$w_i = 0$	$\Delta P(t)$ constant	—	1/2
	$q(t)$ constant	—	4/5
	$q(t) \propto t$	—	7/5
$w_i \neq 0$	$\Delta P(t)$ constant	—	1/2
	$q(t)$ constant	Early	1/2
		Intermediate	$1/2 < \alpha < 4/5$
		Late	4/5
	$q(t) \propto t$	Early	1/2
		Intermediate	$1/2 < \alpha < 7/5$
		Late	7/5

References

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